Fiscal Year:	FY 2006	**Task Last Updated:**	FY 10/24/2005
PI Name:	Wilson, John W. Ph.D.		
Project Title:	Consortium for Shielding Models and Computational Procedures		
Division Name:	Physical Sciences		
Program/Discipline:	RADIATION SHIELDING		
Program/Discipline--Element/Subdiscipline:	RADIATION SHIELDING--Radiation shielding		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Hampton	**State:**	VA
Zip Code:	23681-0001	**Congressional District:**	1
Comments:	srhp.jsc.nasa.gov		
Project Type:	Ground	**Solicitation / Funding Source:**	01-OBPR-05
Start Date:	11/15/2002	**End Date:**	11/14/2006
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	3	**No. of Master' Degrees:**	
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA MSFC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Tripathi, Ram (NASA Langley Research Center) Heinbockel, John (Old Dominion University) Tweed, John (Old Dominion University)		
Grant/Contract No.:	Intramural		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	To systematically treat physical perturbation approach as first order reduction of solutions in terms of continuous slowing down plus straggling followed by treatment of multiple elastic scattering and nuclear reaction. The nuclear reaction portion is reduced to forward and isotropic component. Multiple elastic scattering and the forward nuclear reactive component are treated in asymptotic expansion about forward direction. Two approaches are used to solve forward problem: perturbation expansion and marching procedures. The first three perturbation terms are solved analytically and higher order terms evaluated using non-perturbative methods and used to renormalize third perturbation term to account to all orders. The second asymptotic term is then evaluated as off axis correction. The isotropic source of particles from nuclear reactions will be evaluated by first order perturbation theory and a new isotropic source evaluated and consist mainly of relatively low energy neutrons and are treated by approaches similar to those used in the nuclear power community. A current version of the code will be used to study the sensitivity to nuclear fragmentation cross section error. To develop a high performance 3D Boltzmann equation solver for space and laboratory use and investigate sensitivity of solutions to uncertainties in the nuclear cross sections. Protecting the astronaut from the hazards of space radiation is one of NASA's two highest priorities (O'Keefe, 2002). It is a tall pole in NASA exploration program. This task provides the software used in protecting astronauts in NASA missions and used to augment the ISS shield.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Methods and data used in cancer thearpy
Task Progress:	A number of activities have been carried out under this project during this task period. Besides providing the needed radiation damage technical input for the agency's missions, we have been invited and participated in international meetings/conferences on issues related to radiation damage, in general, and space radiation missions in particular. For example, we were the only US group that was invited to participate in the Benchmark Activity at the 35th COSPAR Scientific Assembly in Paris on the Ground-based Iron-Ion Biology and Physics, including Shielding. The goal was to summarize radiation biology and physics experiments from world-wide facilities providing iron-ion beams for investigations of space radiobiology. A brief synopsis of some of the activities is as follows: • Verification and Validation: High Energy and Charge (HZE) Transport Codes Future Development Formalism for further developing a fully three dimensional HZETRN code using marching procedures but also development of a new code which is capable of not only validation in the space environment but also in ground based laboratories with directed beams of ions of specific energy and characterized with detailed diagnostic particle spectrometer devices. Special emphasis is given to verification of the computational procedures and validation of the resultant computational model using laboratory and spaceflight measurements. At the current state of affairs, parallel development paths for computational model implementation using marching procedures and Green's function techniques are followed as has arisen from historical demands and requirements. A new version of the HZETRN code capable of simulating HZE ions with either laboratory or space boundary conditions is under development. Validation of computational models at this time is particularly important in preparation for design of the Bush Initiative infrastructure for human exploration and demonstration of the Crew Exploration Vehicle in low Earth orbit in 2008. • Space Radiation Transport Methods Development Early space radiation shield code development relied on Monte Carlo methods and made important contributions to the space program. Even so, intensive computational requirements resulted and shield evaluation was made near the end of the design process and resolving shielding issues usually had a negative impact on the design. Improved spacecraft shield design requires early entry of radiation constraints into the design process to maximize performance and minimize costs. As a result, we have been investigating high-speed computational procedures to allow shield analysis from the preliminary design concepts to the final design. In particular, we discuss the progress towards a full three-dimensional and computationally efficient deterministic code for which the current HZETRN evaluates the lowest order asymptotic term. HZETRN is the first deterministic solution to the Boltzmann equation allowing field mapping within the International Space Station (ISS) in tens of minutes using standard Finite Element Method (FEM) geometry common to engineering design practice enabling development of integrated multidisciplinary design optimization methods. A single ray trace in ISS geometry requires 14 milliseconds. A potential means of improving the Monte Carlo efficiency in coupling to spacecraft geometry has been given in terms of reconfigurable computing and could be utilized in the final design as verification of the deterministic method optimized design. • An improved Green's Function Code for Ion Beam Transport. A new Green's function code (GRNTRN) which capable of simulating HZE ions with either laboratory or space boundary conditions is currently under development. The computational model consists of combinations of physical perturbation expansions based on the scales of atomic interaction, multiple scattering, and nuclear reactive processes with use of Neuman-asymptotic expansions with non-perturbative corrections. The code contains energy loss with straggling, nuclear attenuation, nuclear fragmentation with energy dispersion and down shifts. • Cross section sensitivity and propagated errors in HZE exposures It has long been recognized that galactic cosmic rays are of such high energy that they tend to pass through available shielding materials resulting in exposure of astronauts and equipment within space vehicles and habitats. Any protection provided by shielding materials result not so much from stopping such particles but by changing their physical character in interaction with shielding material nuclei forming, hopefully, less dangerous species. Clearly, the fidelity of the nuclear cross-sections is essential to correct specification of shield design and sensitivity to cross-section error is important in guiding experimental validation of cross-section models and database. We examine the Boltzmann transport equation which is used to calculate dose equivalent during solar minimum, with units (cSv/yr), associated with various depths of shielding materials. The dose equivalent is a weighted sum of contributions from neutrons, protons, light ions, medium ions and heavy ions. We investigate the sensitivity of dose equivalent calculations due to errors in nuclear fragmentation cross-sections. We do this error analysis for all possible projectile-fragment combinations (14,365 such combinations) to estimate the sensitivity of the shielding calculations to errors in the nuclear fragmentation cross-sections. Numerical differentiation with respect to the cross-sections is evaluated in a broad class of materials including polyethylene, aluminum and copper. Most important cross-sections for further experimental study are identified and their impact on propagated errors has been evaluated in shielding estimates.

	• Computational Methods for the HZETRN Code Asymptotic expansion has been used to simplify the transport of high charge and energy ions for broad beam applications in the laboratory and space. The solution of the lowest order asymptotic term is then related to a Green's function for energy loss and straggling coupled to nuclear attenuation providing the lowest order term in a rapidly converging Neumann series for which higher order collisions terms are related to the fragmentation events including energy dispersion and downshift. The first and second Neumann corrections were evaluated numerically as a standard for further analytic approximation. The first Neumann correction is accurately evaluated over the saddle point whose width is determined by the energy dispersion and located at the downshifted ion collision energy. Introduction of the first Neumann correction leads to significant simplification of the second correction term allowing application of the mean value theorem and a second saddle point approximation. The regular dependence of the second correction spectral dependence lends hope to simple approximation to higher corrections. At sufficiently high energy nuclear cross section variations are small allowing non-perturbative methods to all orders and renormalization of the second corrections allow accurate evaluation of the full Neumann series.
Bibliography Type:	Description: (Last Updated: 08/20/2012)
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